

$$397. \left(x + \frac{1}{x}\right) = 2 \Rightarrow \left(x + \frac{1}{x}\right)^2 = 2^2$$

$$\Rightarrow x^2 + \frac{1}{x^2} + 2 = 4 \Rightarrow x^2 + \frac{1}{x^2} = 2$$

$$\Rightarrow x^2 + \frac{1}{x^2} - 2 \cdot x \cdot \frac{1}{x} = 2 - 2 = 0$$

$$\Rightarrow \left(x - \frac{1}{x}\right)^2 = 0 \Rightarrow x - \frac{1}{x} = 0.$$

$$398. \left(a + \frac{1}{a}\right) = 6 \Rightarrow \left(a + \frac{1}{a}\right)^2 = 6^2 = 36$$

$$\Rightarrow a^2 + \frac{1}{a^2} + 2 = 36 \Rightarrow \left(a^2 + \frac{1}{a^2}\right) = 34$$

$$\Rightarrow \left(a^2 + \frac{1}{a^2}\right)^2 = (34)^2$$

$$\Rightarrow a^4 + \frac{1}{a^4} + 2 = 1156 \Rightarrow \left(a^4 + \frac{1}{a^4}\right) = 1154.$$

$$399. \left(x - \frac{1}{x}\right) = 2 \Rightarrow \left(x - \frac{1}{x}\right)^2 = 2^2 = 4$$

$$\Rightarrow x^2 + \frac{1}{x^2} - 2 = 4 \Rightarrow x^2 + \frac{1}{x^2} = 6$$

$$\Rightarrow \left(x^2 + \frac{1}{x^2}\right)^2 = 6^2 = 36$$

$$\Rightarrow x^4 + \frac{1}{x^4} + 2 = 36 \Rightarrow x^4 + \frac{1}{x^4} = 34.$$

$$400. \left(x - \frac{1}{x}\right) = \sqrt{21} \Rightarrow \left(x - \frac{1}{x}\right)^2 = (\sqrt{21})^2 = 21$$

$$\Rightarrow x^2 + \frac{1}{x^2} - 2 = 21 \Rightarrow x^2 + \frac{1}{x^2} = 23$$

$$\Rightarrow x^2 + \frac{1}{x^2} + 2 = 25$$

$$\Rightarrow \left(x + \frac{1}{x}\right)^2 = 5^2 \Rightarrow x + \frac{1}{x} = 5.$$

$$\therefore \left(x^2 + \frac{1}{x^2}\right) \left(x + \frac{1}{x}\right) = 23 \times 5 = 115.$$

$$401. x^4 + \frac{1}{x^4} = 322 \Rightarrow x^4 + \frac{1}{x^4} + 2 = 324$$

$$\Rightarrow \left(x^2 + \frac{1}{x^2}\right)^2 = (18)^2 \Rightarrow x^2 + \frac{1}{x^2} = 18$$

$$\Rightarrow x^2 + \frac{1}{x^2} - 2 = 16$$

$$\Rightarrow \left(x - \frac{1}{x}\right)^2 = 16 \Rightarrow \left(x - \frac{1}{x}\right) = 4.$$

$$402. \left(x + \frac{1}{x}\right) = p \Rightarrow \left(x + \frac{1}{x}\right)^3 = p^3$$

$$\Rightarrow x^3 + \frac{1}{x^3} + 3 \cdot x \cdot \frac{1}{x} \left(x + \frac{1}{x}\right) = p^3$$

$$\Rightarrow x^3 + \frac{1}{x^3} + 3p = p^3 \Rightarrow x^3 + \frac{1}{x^3} = p^3 - 3p.$$

$$403. \left(x - \frac{1}{x}\right) = 1 \Rightarrow \left(x - \frac{1}{x}\right)^3 = 1$$

$$\Rightarrow x^3 - \frac{1}{x^3} - 3 \cdot x \cdot \frac{1}{x} \left(x - \frac{1}{x}\right) = 1$$

$$\Rightarrow x^3 - \frac{1}{x^3} - 3 = 1 \Rightarrow x^3 - \frac{1}{x^3} = 1 + 3 = 4.$$

$$404. \left(a^4 + \frac{1}{a^4}\right) = 1154 \Rightarrow a^4 + \frac{1}{a^4} + 2 = 1156$$

$$\Rightarrow \left(a^2 + \frac{1}{a^2}\right)^2 = 1156 \Rightarrow \left(a^2 + \frac{1}{a^2}\right) = 34$$

$$\Rightarrow a^2 + \frac{1}{a^2} + 2 = 36$$

$$\Rightarrow \left(a + \frac{1}{a}\right)^2 = 36 \Rightarrow a + \frac{1}{a} = 6$$

$$\Rightarrow \left(a + \frac{1}{a}\right)^3 = 6^3 = 216$$

$$\Rightarrow a^3 + \frac{1}{a^3} + 3 \cdot a \cdot \frac{1}{a} \left(a + \frac{1}{a}\right) = 216$$

$$\Rightarrow a^3 + \frac{1}{a^3} + 3 \times 6 = 216$$

$$\Rightarrow a^3 + \frac{1}{a^3} = 216 - 18 = 198.$$

$$405. x + \frac{1}{x} = 3 \Rightarrow \left(x + \frac{1}{x}\right)^3 = 3^3 = 27$$

$$\Rightarrow x^3 + \frac{1}{x^3} + 3 \cdot x \cdot \frac{1}{x} \left(x + \frac{1}{x}\right) = 27$$

$$\Rightarrow x^3 + \frac{1}{x^3} + 3 \times 3 = 27 \Rightarrow x^3 + \frac{1}{x^3} = 18$$

$$\Rightarrow \left(x^3 + \frac{1}{x^3}\right)^2 = (18)^2 = 324$$

$$\Rightarrow x^6 + \frac{1}{x^6} + 2 = 324 \Rightarrow x^6 + \frac{1}{x^6} = 322.$$

$$406. x + \frac{1}{x} = \sqrt{13} \Rightarrow \left(x + \frac{1}{x}\right)^2 - 4 = (\sqrt{13})^2 - 4 = 13 - 4 = 9$$

$$\Rightarrow \left(x - \frac{1}{x}\right)^2 = 9$$

$$\Rightarrow \left(x - \frac{1}{x}\right) = 3$$

$$\Rightarrow \left(x - \frac{1}{x}\right)^3 = 3^3 = 27$$

$$\Rightarrow x^3 - \frac{1}{x^3} - 3 \cdot x \cdot \frac{1}{x} \left(x - \frac{1}{x}\right) = 27$$

$$\Rightarrow x^3 - \frac{1}{x^3} - 3 \times 3 = 27 \Rightarrow x^3 - \frac{1}{x^3} = 27 + 9 = 36.$$